

Nirma University
Institute of Technology
Semester End Examination (IR), December - 2025
Master of Computer Applications, Semester - I
6CS105CC22: Computer Networks

Roll/
Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

- Instructions:
1. **All questions are compulsory. (No Optional Questions)**
 2. **Use a section-wise separate answer book.**
 3. Figures to the right indicate full marks.
 4. Draw neat sketches wherever necessary.
 5. Assume suitable data wherever necessary and mention the same.

Section-I

Q-1 : Answer the following questions. [16]

- CO1 (a) Explain the key functions of Application layer and Data Link layer of OSI (Open Systems Interconnection) Reference Model. (8)
BL3
CO1 (b) Distinguish between a port address, a logical address, and a physical address by providing the role of these addresses. Also discuss the protocol layers in which these addresses are applicable. (8)
BL2

Q-2 : Answer the following questions. [18]

- CO1 (a) Why controlled access protocols are better than random access protocols? Explain with proper justifications. Also, discuss any one controlled access protocol using a suitable diagram. (6)
BL3
CO1 (b) Why is Framing required at the data link layer? Give suitable reasons to justify your answer. Also, explain any three header fields of a data link layer frame. (6)
BL2
CO2 (c) Explain using a suitable example, how the Code-division multiple access method works. (6)
BL3

Q-3 : Answer the following questions. [16]

- CO2 (a) Assume a 7-bit data word as 1110000. Using the hamming code (non-linear) Error detection and correction method, generate all required check bits and find the corresponding hamming code word for the above-mentioned data word. Now, when a single-bit error is introduced at the second MSB bit then how is the error detected using the hamming code method? Discuss it. (8)
BL3
CO3 (b) Fill in the entries in the following router table (R1) under the Network address column having "?" value. Thereafter, draw the network diagram for the network using the routing table of router R1. (8)
BL3

Mask	One of the Address	Network Address	Next-Hop	N/W Address Interface
/26	199.14.17.20	?	-----	M1
/19	150.23.192.1	?	-----	M0
/21	177.25.64.100	?	-----	M3
Default	Default	Default	156.90.12.4	M2

Section - II**Q-4****Answer the following questions.****[16]**CO3
BL2

- (a) Explain the concept of a Virtual Private Network (VPN) and its role in ensuring secure communication over public networks. Discuss the key components and protocols involved in VPNs (IPsec, OpenVPN) and differentiate between various types of VPNs. (8)

CO3
BL3

- (b) Explain the following ICMP (Internet Control Message Protocol) Error reporting messages in detail by taking a suitable example: (8)
- i) Source quench ii) Time exceeded

Q-5**Answer the following questions:****[16]**CO3
BL3

- (a) One of the IPv4 addresses in a block is 170.222.170.83/27. Using this data, find the network mask, the number of host addresses in the network, the first host address, and the last address in this network. (8)

CO2
BL2

- (b) Explain the working mechanism of the Domain Name System (DNS) and its role in Internet communication. Discuss the different types of DNS servers and records, their functionalities, and how DNS caching improves performance. (8)

Q-6**Answer the following questions.****[18]**CO3
BL2

- (a) Illustrate the TCP handshaking process to start the data segment and acknowledgment transfer process between two communicating end systems using a timeline diagram. (10)

Also, explain the error control mechanisms employed in the following cases for TCP segments:

- i) When the segment is lost
ii) Fast retransmission
iii) When the acknowledgment is lost

CO3
BL4

- (b) An IP router with a Maximum Transmission Unit (MTU) of 1200 bytes has received an IP packet of size 2204 bytes with an IP header of length 20 bytes. To send the IP packet the Router will fragment the packet. For this case, answer the following questions: (8)
- i) How many fragments would reach the destination?
ii) For each fragment, find out the value of the More Fragments bits, Fragment Offset, Data length, and the Total length of the fragments.